

ResQttech: A Computer Vision Based Risk Level Detection and Dispatch-Estimation Framework for UAV-Assisted Flood Rescue Operations

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Abstract—Larger-scale floods generally overwhelm manual rescue coordination that depends on phone calls and manually updated maps that make a prediction on how many people might be at risk and how many rescuers need to be deployed. This paper describes a framework named resQttech that uses computer vision techniques to turn the task of flood victim identification into a risk stratification based detection and dispatch estimation problem. The risk classes are: Critical, High, Medium, and Low Risk or Rescue Individual. Using a YOLOv12n detector on a custom dataset for flood scenes, we perform classification of individuals into one of these risk categories. The ResQttech Dispatch Algorithm uses a set of rules that translates per frame counts of individuals from each category into a number of rescuers to be deployed and an alert level. With 7,460 annotated images in the dataset, our detector achieves precision of 0.918, recall of 0.908, mAP@0.5 of 0.934, and mAP@0.5:0.95 of 0.583. We test the performance of YOLOv5s, YOLOv8n, YOLOv9s, YOLOv10n, YOLOv11n, and YOLOv12n on the same dataset and find out that YOLOv12n performs the best in terms of mAP@0.5 (0.934) while YOLOv8n provides the best accuracy/performance ratio.

Index Terms—Flood, YOLOv12, Risk-Aware Object Detection, Search and Rescue, Dispatch Estimation, Disaster Response, Computer Vision

I. INTRODUCTION

In August 2024, the districts of Cumilla and Feni in Bangladesh experienced severe floods. Within hours, roads became impassable and visibility dropped, overwhelming responders who lacked precise information about trapped residents. Rescue coordination currently relies on fragmented, delayed data that can cost lives. While aerial camera footage is increasingly available, manual review does not scale during large events. Even automated systems typically stop at binary detection: a person is flagged as present, but no judgment is made about relative urgency, and no concrete estimate is produced regarding resource deployment. Instead of considering victim detection as a binary process, resQttech categorizes each detected victim into one of five different risk levels and adds a dispatch estimation layer that translates category-specific counts to a practical number of rescue teams to be dispatched. Our innovations lie in recasting the flood-

victim detection problem as a five-class, risk-aware detection task. A custom-annotated image dataset consisting of 7,460 images; Training of YOLOv12n for establishing benchmarking metrics; Comparison of 6 architectures of YOLO (v5s to v12n); and the creation of a ResQttech Dispatch Algorithm (RDA). Furthermore, we deployed *Shohay* a command center web interface that utilizes our framework to provide real-time coordination for disaster response.

II. RELATED WORK

UAV-assisted human detection for disaster response heavily leverages lightweight deep-learning models to achieve real-time performance [1], [2]. Aerial imagery analysis has been significantly advanced by convolutional networks [4], [9], U-Net architectures for flood-mapping [5], and transformer-based approaches [3]. Comprehensive surveys underscore the transformative impact of machine learning in UAV navigation, data collection, and remote sensing for disaster assessment [7], [8], [6]. For real-time monitoring, multimodal data fusion tracks landmass and water levels [10], while generative AI addresses data scarcity by synthesizing training imagery for YOLO-based models [11].

Search and rescue (SAR) missions require high accuracy under demanding conditions. YOLO-based architectures are widely adopted; for instance, RescueNet utilizes a modified YOLO model to count flood survivors [12], and YOLOv3 has been applied for combined human detection and action recognition [13]. To handle the computational constraints of UAVs, researchers have proposed collaborative architectures utilizing lightweight models (e.g., Ghost Modules and MobileNet blocks) combined with reinforcement learning for intelligent task offloading [14]. Additional studies demonstrate that YOLO algorithms coupled with tracking models like Deep SORT enhance human detection in dynamic disaster zones [15]. Addressing domain-specific challenges, recent works focus on fine-tuning models on niche datasets (e.g., people trapped in buildings) [16], detecting stranded individuals in flooded environments using enhanced multi-scale architectures

like H-YOLO [17], and specifically adapting models (e.g., YoloOW) for open water environments where scale variations and water surface flares cause false detections [18], [19], [20].

Despite these advances, most systems stop at binary detection without translating visual data into urgency-stratified, operational response recommendations. resQtech bridges this gap by coupling risk-stratified classification directly with a dispatch-estimation layer.

III. DATASET

The data set employed for this study has been compiled exclusively for object detection in flood affected areas. The data set includes five categories of objects that have different semantics. These include **Critical Risk** (immediate risk to life), **High Risk** (severe risks necessitating immediate action), **Medium Risk** (distress), **Low Risk** (individuals who are stable and need to be observed) and **Rescue Individual** (people at the scene that are important to recognize so that victims are not mistakenly identified). The dataset made of 7,460 annotated images sourced from the Roboflow Universe platform. It is partitioned into three non-overlapping splits as shown in Table I.

TABLE I
DATASET PARTITION SPLITS

Split	Images	Proportion
Training	6,525	~87.5%
Validation	465	~6.2%
Test (held-out)	470	~6.3%
Total	7,460	100%

Bounding box annotation was carried out for each object in the image. Class imbalance in the data set is present to some extent. *Critical Risk* and *Low Risk* are the most commonly annotated classes, covering about 57% of all the bounding boxes. This is indicative of the reality that high or low severity cases in disasters can be more easily detected than intermediate severity cases.

TABLE II
PER-CLASS BOUNDING BOX INSTANCE COUNTS ACROSS DATASET SPLITS.

Class	Train	Valid	Test	Total
Critical Risk	2,041	162	144	2,347
High Risk	1,557	97	151	1,805
Low Risk	2,049	134	111	2,294
Medium Risk	504	46	42	592
Rescue Individual	936	60	60	1,056
Total Instances	7,087	499	508	8,094

A. Preprocessing and Augmentation

Prior to training, all images were auto-oriented (EXIF data removal) and resized to 640×640 pixels using stretch interpolation. In order to improve model generalization, the augmentation procedure was performed offline and created three augmented images from each source image, by applying a 50% probability of horizontal flip, and 0.3% of salt and pepper noise. The YOLOv12 model training process includes an additional online HSV color jittering (hue ± 0.015 , saturation ± 0.7 , value ± 0.4), scale/translation jittering, and mosaic augmentation (excluded for the last 10 epochs).

IV. METHODOLOGY

The resQtech process takes place in two steps: (1) detection of individuals by a YOLOv12n detector into five categories, and (2) conversion of per frame counts to a dispatch algorithm through ResQtech Dispatch Algorithm.

A. ResQtech Dispatch Algorithm (RDA)

In order to close the divide between mere detection and practical application, the ResQtech Dispatch Algorithm (RDA) was created. The rule-based algorithm analyzes the output of the bounding boxes generated by the YOLO detector in order to estimate the real-time requirement for rescuers.

This algorithm uses a diminishing return scaling factor (square root) for *Critical* and *High* threat classes to avoid an exponential explosion during overcrowded scenes, using a regular linear scaling factor for *Medium* and *Low* risk classes. Most importantly, the algorithm calculates the total dispatch requirement by subtracting the number of *Rescue Individuals* that are already present in the scene.

The core dispatch logic is defined as:

$$N_{req} = \lceil 2.0\sqrt{C_{crit}} \rceil + \lceil 1.0\sqrt{C_{high}} \rceil + \lceil 0.5C_{med} \rceil + \lceil 0.25C_{low} \rceil \quad (1)$$

$$D_{final} = \max(0, \min(N_{req} - C_{rescuer}, 20)) \quad (2)$$

Where C represents the detected count of each respective class, N_{req} is the total required personnel, and D_{final} is the final dispatch count capped at a maximum of 20 responders per scene. The algorithm also assigns a dynamic scene alert level based on the highest risk category present. Figure 1 summarizes the entire processing pipeline from detection to the final dispatch report.

V. EXPERIMENTAL SETUP

Main detector (YOLOv12n) was trained for 30 epochs, batch size 10, image size 640×640 , on CPU, with early-stopping patience of 5. The optimizer was auto-selected (lr0 0.01, momentum 0.937, weight decay 0.0005) with AMP enabled. Validation used IoU 0.7 for NMS.

VI. RESULT ANALYSIS

Detection accuracies for our main YOLOv12 model are shown in the following tables. In custom dataset containing flood-scenes, the best values obtained for mean Average Precision (mAP) are 0.934 at mAP@0.5 and 0.583 at mAP@0.5:0.95 within 30 epochs

A. Class-wise Detection Performance

This research evaluates the performance of YOLOv12 across five distinct risk categories to analyze detection accuracy in complex disaster environments. After comparing the detection results across the risk levels, the mAP@0.5 score is consistently high across all classes, indicating robust detection accuracy. *Medium Risk* achieved the highest mAP@0.5 (0.948), while *Rescue Individual* scored slightly lower (0.919). This minor drop is primarily due to occasional visual overlap with victims in cluttered flood waters. Table III confirms that

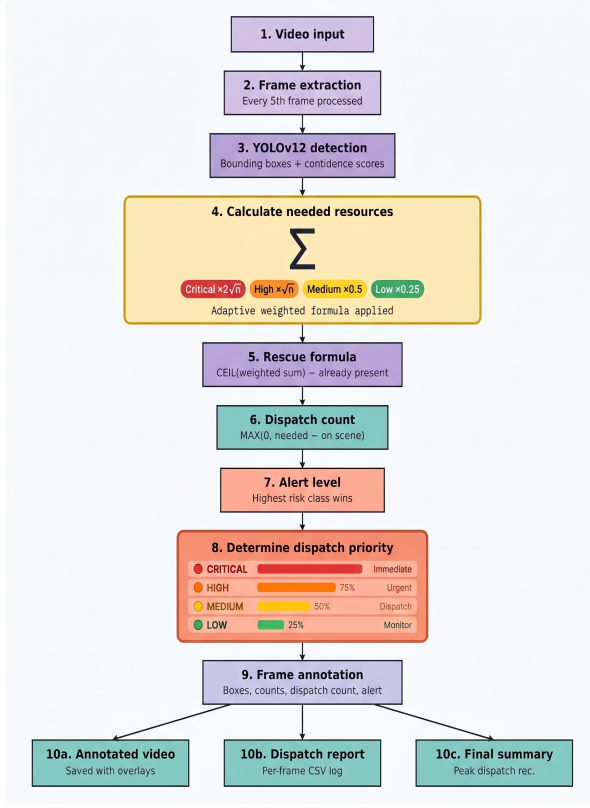


Fig. 1. resQtech processing pipeline: detection feeds the ResQtech Dispatch Algorithm, producing per-frame annotations, a dispatch report, and a peak-dispatch summary.

the YOLOv12 architecture effectively discriminates between severity levels.

TABLE III
CLASS-WISE DETECTION RESULTS USING YOLOV12.

Category	Instances	Precision	Recall	mAP@0.5	mAP@0.5:0.95
All Classes	1,044	0.918	0.908	0.934	0.583
Critical Risk	298	0.937	0.940	0.928	0.551
High Risk	183	0.945	0.950	0.927	0.562
Low Risk	222	0.925	0.930	0.936	0.590
Medium Risk	178	0.945	0.940	0.948	0.605
Rescue Individual	165	0.946	0.950	0.919	0.574

The classification performance is highly accurate, with the diagonal showing strong true-positive rates. The most notable errors are background false negatives (e.g., 28 *critical risk* instances missed as background), which is expected given the extreme visual noise, floating debris, and poor lighting inherent in disaster imagery. Cross-class confusion among the risk categories remains remarkably low. In Figure 2, the confusion matrix for the dataset is presented.

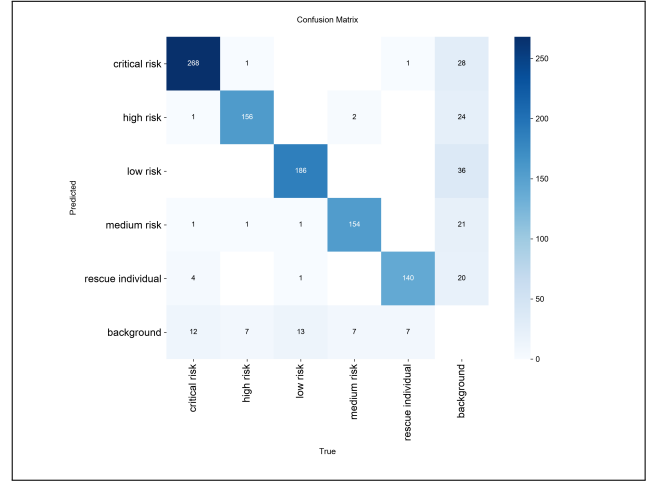


Fig. 2. Confusion matrix of YOLOv12 on the flood dataset.

B. Comparison of YOLO Models

While earlier architectures like YOLOv5s and YOLOv9s demonstrate strong baseline performance, YOLOv12n clearly outperforms the other models across all metrics, achieving a peak mAP@0.5 of 0.934. This indicates that given adequate training cycles, YOLOv12's advanced spatial attention mechanisms are highly suited for identifying small, occluded human subjects in chaotic flood environments. In Table IV, we present a comparison of multiple YOLO models (v5s, v8n, v9s, v10n, v11n, and v12n).

TABLE IV
BENCHMARK OF YOLO ARCHITECTURES (10 EPOCHS).

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv5s	0.877	0.837	0.889	0.517
YOLOv8n	0.874	0.798	0.875	0.511
YOLOv9s	0.871	0.807	0.886	0.520
YOLOv10n	0.875	0.819	0.884	0.516
YOLOv11n	0.843	0.836	0.883	0.515
YOLOv12n (30 ep.)	0.918	0.908	0.934	0.583

C. Confidence Curves and Training Results

A peak overall F1 score of 0.91 is achieved at a confidence threshold of 0.438. Figure 3 illustrates the F1 score of our model, highlighting its ability to balance precision and recall.

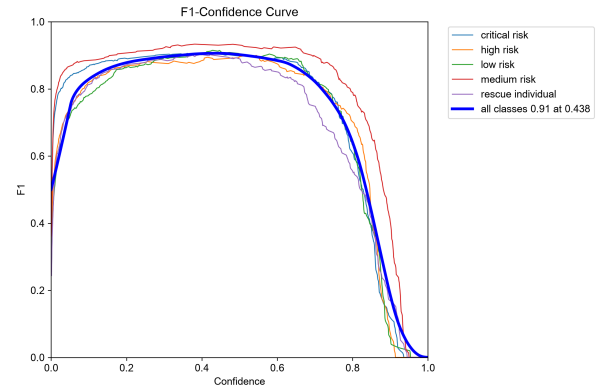


Fig. 3. F1 Curve of YOLOv12 on the resQtech dataset.

Demonstrating an excellent trade-off between precision and recall across all risk categories, with the curve hugging the top-right corner, Figure 4 presents the Precision-Recall curve.

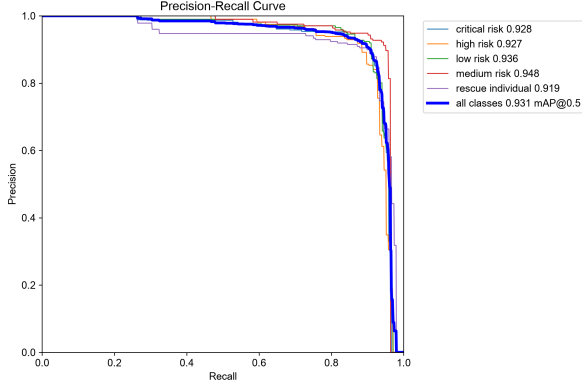


Fig. 4. Precision-Recall Curve of YOLOv12.

The training losses steadily decrease, while the validation metrics stabilize and plateau after approximately epoch 20, indicating stable learning without significant overfitting. Figure 5 depicts our model's loss and metric convergence over 30 epochs.

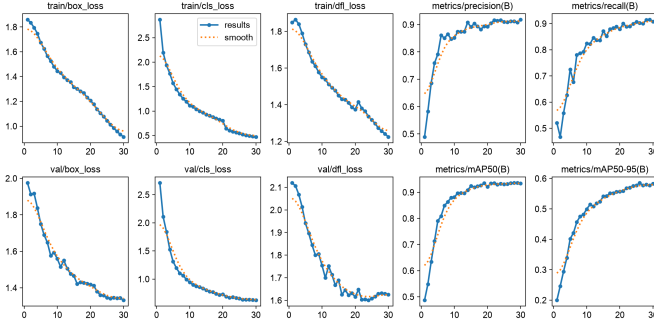


Fig. 5. Detection metrics and loss graphs over 30 epochs for YOLOv12.

This illustrates the diverse environmental conditions, bounding box placements, and real-time augmentations (such as mosaic and color jitter) applied during the YOLOv12 training process to enhance model robustness. Figure 6 provides a visualization of a sample training batch.

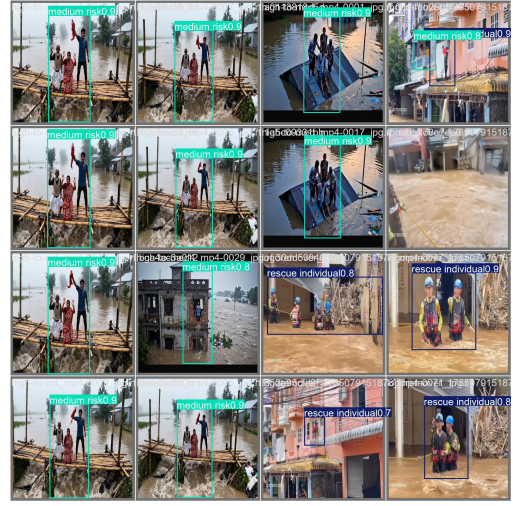


Fig. 6. Training dataset images used for YOLOv12.

The outputs confirm that the model successfully isolates victims and rescue personnel, accurately classifying their respective risk levels even in highly occluded and visually challenging scenarios. Finally, Figure 7 visualizes the model's bounding box outputs on unseen test data.



Fig. 7. Test dataset labels and predictions using YOLOv12.

D. Evaluation on Held-out Test Dataset

To validate the model's generalization capabilities, we evaluated the trained YOLOv12 architecture on a strictly held-out test split comprising over 500 unseen images. The model generated a total of 1,387 distinct bounding box predictions across the five risk categories. The test results demonstrate that the model actively identifies subjects across all severity levels without heavy bias toward a single class. Notably, the model exhibits its highest average confidence when detecting *Critical Risk* (0.730) and *Rescue Individual* (0.727) instances. This is an operationally ideal outcome, as these two classes are the most crucial for the ResQtch Dispatch Algorithm to accurately estimate immediate resource needs and account for personnel already deployed in the water. Table V details

the distribution of these detections and the model’s average confidence score for each category.

TABLE V
DETECTION DISTRIBUTION AND CONFIDENCE ON UNSEEN TEST DATA.

Category	Total Detections	Average Confidence
Critical Risk	280	0.730
High Risk	276	0.698
Low Risk	299	0.682
Medium Risk	358	0.671
Rescue Individual	174	0.727
All Classes	1,387	0.702

E. Dispatch Estimation Results

For performance analysis of the ResQtech Dispatch Algorithm, we executed the end-to-end YOLOv12+RDA pipeline on different test videos featuring flood-related rescue scenes. The processing is done by the algorithm on a per-video-frame basis, with the rescuer dispatch count rendered on the top left corner of each video frame, as illustrated in Figure 8.

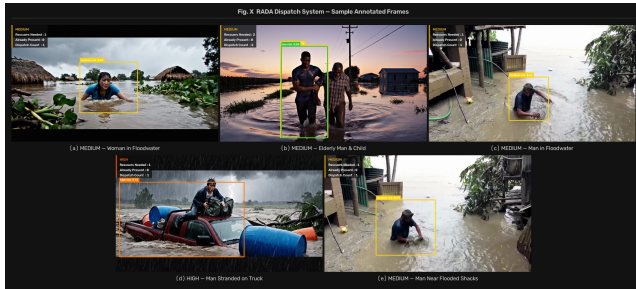


Fig. 8. RDA system interface showing real-time dispatch calculation, subtracting present rescuers from the total requirement.

The results show how versatile the algorithm is. As an example, in "People rescuing person", even though there is a risk involved, the highest number of dispatches needed is 0. This clearly shows how the algorithm works using the concept of subtraction by understanding that there are enough *Rescue Individuals* in the frame that can manage the risk without any additional dispatch. On the other hand, the "Final Compilation" video gives out a "CRITICAL" warning with the highest dispatch needed being 4. Table VI summarizes the algorithm's performance across several multi-frame video clips.

TABLE VI
RDA DISPATCH SUMMARY ON TEST VIDEO CLIPS

Video Scenario	Frames	Peak	Avg	Alert Level
Man saving child	20	2	2	HIGH
Man stranded on roof	20	1	1	MEDIUM
Man stranded on truck	20	1	1	HIGH
Man wading in floodwater	20	2	1	HIGH
People rescuing person	20	0	0	HIGH
Woman swimming in flood	20	3	1	CRITICAL
Young man helping elderly	20	2	2	MEDIUM
Final Compilation	140	4	1	CRITICAL

VII. SYSTEM DEPLOYMENT: SHOHAY WEB INTERFACE

To bridge the gap between algorithmic detection and real-world disaster management, the resQtech framework is integrated into a fully operational web application named *Shohay*.

Serving as a centralized Command Center, Shohay is designed to coordinate disaster response operations, manage citizen assistance requests, and streamline volunteer deployment.

As illustrated in Fig. 9, the dashboard aggregates critical real-time telemetry essential for district coordinators. Key operational modules include tracking open assistance requests, monitoring registered volunteer availability, and logging household relief distributions. The interface also visualizes dynamic data, such as 7-day request trends and district-wise shelter occupancy (e.g., Sunamganj, Sirajganj, Feni). Crucially, it maps active flood alerts by severity level (Critical, High, Medium, Low) which directly aligns with the risk-stratified outputs of our YOLOv12 detector. By coupling the ResQtech Dispatch Algorithm (RDA) with this command center, Shohay ensures that automated deployment recommendations become immediately actionable for partner organizations on the ground.

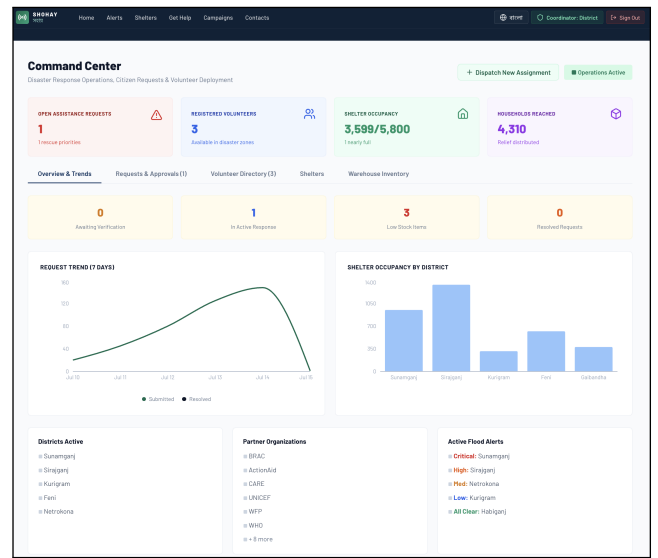


Fig. 9. The Shohay Command Center dashboard, displaying real-time disaster response operations, active flood alerts, and shelter occupancy metrics.

VIII. LIMITATIONS AND FUTURE WORK

Current limitations include CPU-only evaluation without live UAV or thermal integration, and RDA scaling constants remain unvalidated against operational rescue data. Cross-class confusion with the *rescue individual* category directly impacts the dispatch algorithm. Future work will focus on on-board edge deployment, real-time FPS benchmarking, thermal fusion, GPS localization, empirical dispatch calibration, and targeted dataset expansion for rescue personnel.

IX. CONCLUSION

This paper presented resQtech, combining a YOLOv12n five-class risk detector with a rule-based ResQtech Dispatch Algorithm. On 7,460 images, the detector achieved mAP@0.5 of 0.934. The dispatch algorithm demonstrated how risk-category detections can be converted into actionable deployment metrics and alert levels, providing a vital framework for future live UAV integration and automated disaster response.

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